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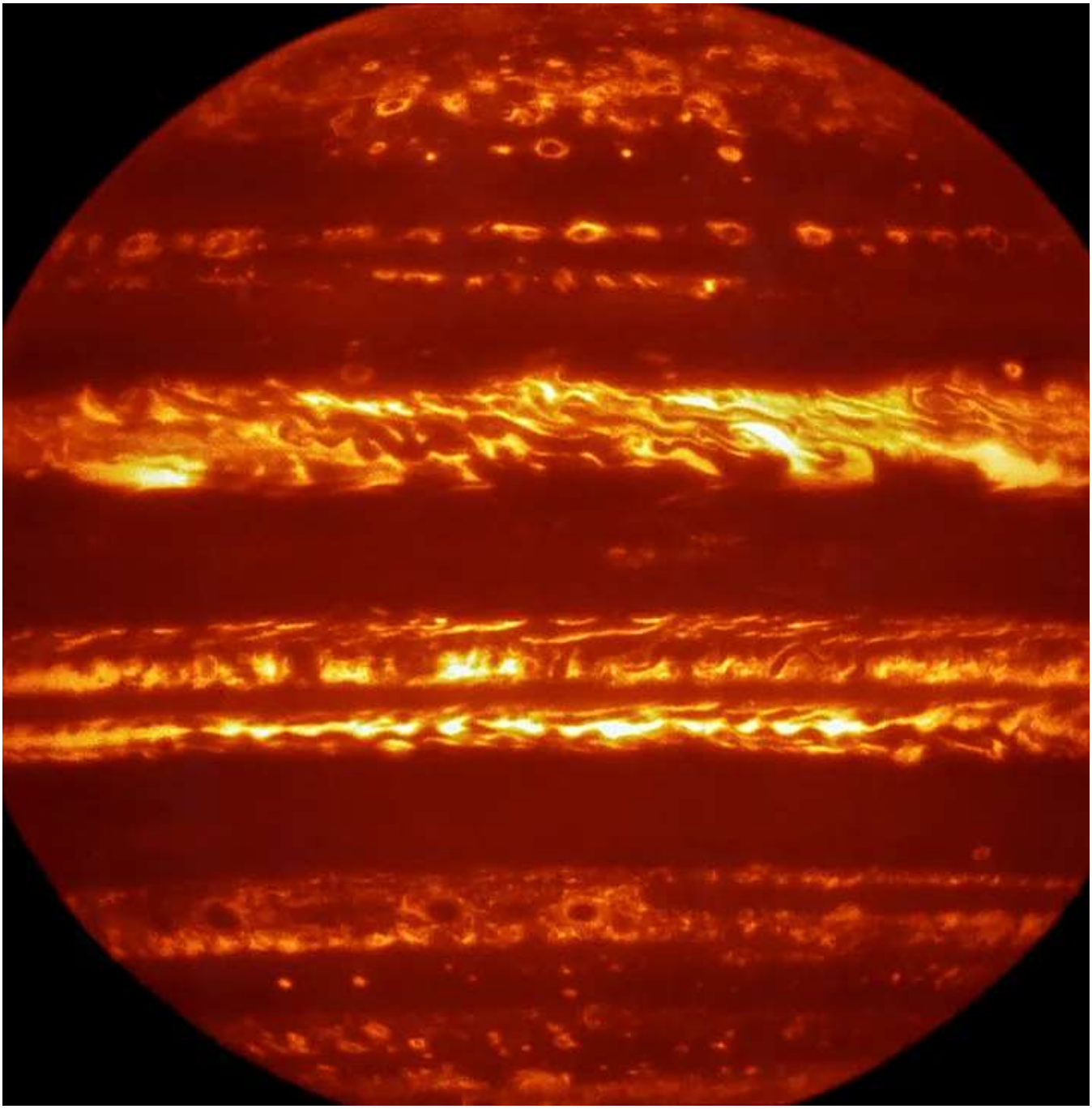
SPACE SCIENCE WITH PYTHON

Space Science with Python — A Rendezvous with Jupiter

Part 9 of the tutorial series focuses on the periodic comets. How are they related to Jupiter and can we see it in the data?

9



Jupiter in very unusual colours. This so called false colour image is an infrared capture of the Solar System's largest planet. It was taken by the European Southern Observatory's (ESO) Very Large Telescope (VLT). The wavelength of the captured light: 5 micrometers (5000 nano-meters). Credit: ESO/L. Fletcher; License: Creative Commons Attribution 4.0 International License

Preface

This is the 9th part of my Python tutorial series "Space Science with Python". All codes that are shown here are uploaded on GitHub. Enjoy!

This tutorial's database has been created in a previous session, but it is also uploaded on my GitHub repository.

Introduction

You do not need a space-bound, billion dollar instrument to do science. Sometimes, a 50 cm telescope (diameter) is enough to impact tremendously the scientific community. It's 1993. Three astronomers analyse their observation findings: Carolyn and Eugene M. Shoemaker and David Levy just discovered a new comet! Its name: Shoemaker-Levy 9.

Shoemaker-Levy 9 is a common comet with a core of 4 km diameter and a typical icy composition. The dynamics however are strange and indicate a cosmic catastrophe, namely the collision of the comet and the largest planet in our Solar System: Jupiter.

A year later the comet approaches Jupiter closer and closer. The tidal forces tear the comet apart in 21 fragments in different sizes. The resulting cosmic *pearl necklace* impacts the planet leaving temporary scars behind. Such an impact would be devastating for our home planet, but Jupiter's gravitational pull is so large that astronomers assume that this planet protects us from any larger debris.

Image composition of Shoemaker-Levy 9's fragments with Jupiter. The black dot on Jupiter is the shadow of one of Jupiter's moons (one can see it at the top right w.r.t. to the shadow). The images were taken with the Hubble Space Telescope. Credit: NASA, ESA, H. Weaver and E. Smith STScI and J. Trauger and R. Evans NASA Jet Propulsion Laboratory

But does Jupiter absorb all objects? Spacecraft missions use the planets for so-called swing-by manoeuvrers to alter their trajectory and velocity. *Do planets, like Jupiter, also alter the orbit of celestial bodies? And do we see this effect in the data?* Let's find out.

Tisserand Parameter

Data science is awesome! State of the art algorithms and methods allow us to derive insights from complex data. It allows us to reduce dimensionality to identify structures and connections. So let's apply these methods on the comets' orbital elements, right?

Working with data, either as a data scientist or as a scientist in academia, requires data engineering. And part of it is feature engineering. In Space Science and especially in astrodynamics sophisticated theories are available that create conceptual fundaments for miscellaneous topics. These theories help us to describe and predict certain aspects; they need to be falsifiable; and they may provide ideas that help us to apply methods with a certain purpose instead of applying blindly some brute force methods. This requires literature research.

Our problem is a dynamical problem. We want to see whether there is a quantitative parameter that helps us to understand a dynamical link between a planet and another object.

Thanks to *Murray & Dermott* there is **the** book for answering any dynamical question: *Solar System Dynamics* [1]. They summarise nicely the ideas of Félix Tisserand [2], who derived in 1896 the theory that orbital elements of a celestial object are "preserved" in a way after a close encounter with a larger planet. Let's assume a comet has a semi-major axis a, an eccentricity e and an inclination i before a close encounter with Jupiter. After a swing-by, the elements are altered to $\tilde{a}$, $\tilde{e}$ and $\tilde{\imath}$. Tisserand found that the following equation must be fulfilled:

$$\frac{1}{2a} + \cos(i)\sqrt{a\,(1-e^2)} = \frac{1}{2\tilde{a}} + \cos(\tilde{\imath})\sqrt{\tilde{a}\,(1-\tilde{e}^2)}$$

Consequently, orbital elements cannot change randomly after a swing-by, but follow a certain constant law that preserves a constant factor.

Further, Tisserand found the later called Tisserand Parameter *T*. Using the same orbital elements and the semi-major axis of e.g. Jupiter (Jup), one can compute the dimensionless parameter as:

$$T_{\mathrm{Jup}} = \frac{a_{\mathrm{Jup}}}{a} + 2\cos(i)\sqrt{\frac{a}{a_{\mathrm{Jup}}}(1-e^2)}$$

What does this parameter mean quantitatively? One major idea (see figure 7 in [3]): A Tisserand parameter between 2 and 3 indicates a so-called Jupiter Family Comet (JFC). A comet sub-type, whose original orbit was altered by Jupiter in a way that it revolves around the Sun in a closer orbit.

Does this apply for some P type comets? Last time, we have seen that most aphelion values for P types were at around 5 AU (approximate semi-major axis of Jupiter).

Python walk

We can answer this question by using Python. First, we need our mandatory libraries that have been used in the last couple of times. This time, we will again create a Kernel Density Estimator (KDE) based distribution of the resulting Tisserand parameter. For this, we need the *scipy* module (line 13).

Part 1/9

Line 3 connects to our already known comet database that was created in tutorial session 7. An SQLite cursor (line 6) is needed later. Line 11 to 13 and 17 to 19 extract the semi-major axis, inclination and eccentricities of P and C type comets, respectively. Further, the corresponding comet's name is used later and we consider only bound orbits with an eccentricity <1.

Part 2/9

For the Tisserand parameter we need the semi-major axis of Jupiter. We use some SPICE functions from the previous sessions to calculate this parameter. Our meta file kernel in line 6 is loaded via the *furnsh* function and contains the path to the spk

kernel for the Solar System bodies and a path to the planetary constants kernel. We set any Ephemeris time (ET) in line 9 by using the *utc2et* function and compute the corresponding Jupiter state vector in ECLIPJ2000 coordinates in line 13 to 16. The observer is the Sun (NAIF code 10) and the target object is the barycentre of Jupiter (NAIF code 5). No aberration effects need to be considered, thus we use the function *spkgeo*. To convert the state vector to orbital elements we use the function *oscltx*, however before (line 19 and 20) we need to extract the G*M value of the Sun from the planetary kernel using the function *bodvcd*. The computed semi-major axis is assigned in line 26 and converted to AU in line 29 (using the function convrt).

Part 3/9

Now we can add a new column in the C and P type dataframes that contains the Tisserand parameter for each individual comet. Line 3 and 4 define the parameter as a lambda function. One needs to take care of converting the inclination values from degrees to radians (line 11 and line 16).

Part 4/9

Let's have a look at the resulting statistics. Line 7 to 10 compute the percentage of JFCs by using our newly computed Tisserand parameter (line 7 and 8).

```
Descriptive statistics of the Tisserand parameter of P type comets
count    627.000000
mean       2.672696
std        0.555275
min       -0.648993
25%        2.610908
50%        2.805879
75%        2.935148
max        3.663401
Name: TISSERAND_JUP, dtype: float64
```

According to our computation 83 % of all P type comets are JFCs and thus dynamically linked to Jupiter! *Is this a lot?* Let's take a look at the …

```
Percentage of P type comets with a Tisserand parameter between 2 and
3: 83.0%
```

… C type comets. 75 % of the comets have a Tisserand parameter smaller than 1.35. Small Tisserand parameters are associated with highly eccentric orbits with large semi-major axis. Well … this applies perfectly for our long period comets with an orbit time of more than 200 years.

```
Descriptive statistics of the Tisserand parameter of C type comets
count    159.000000
mean       0.535408
std        1.365811
min       -2.758114
25%       -0.412254
50%        0.548308
75%        1.357792
max        3.578201
Name: TISSERAND_JUP, dtype: float64
```

The newly computed parameter should be stored in the database. *But how can we add a new column to an already existing table?* This code snippet will help us. A function is defined (and will be used in future tutorials) that extracts the column names (line 30) with the SQL PRAGMA command. If the column name exists (line 33), nothing happens. Otherwise (line 37 to 40) a new column with the requested

SQL column type is added. In line 43 to 47 we add the Tisserand parameter as a new column. Please note: Executing this part will have no effect since you will download also the updated comet database from my GitHub repository.

Part 6/9

The command *executemany* enables us to e.g., update a table with several rows of data. In line 2+3 and 6+7 we update the Tisserand parameter, where we use the comet's name as a key between the SQL table and the dataframe.

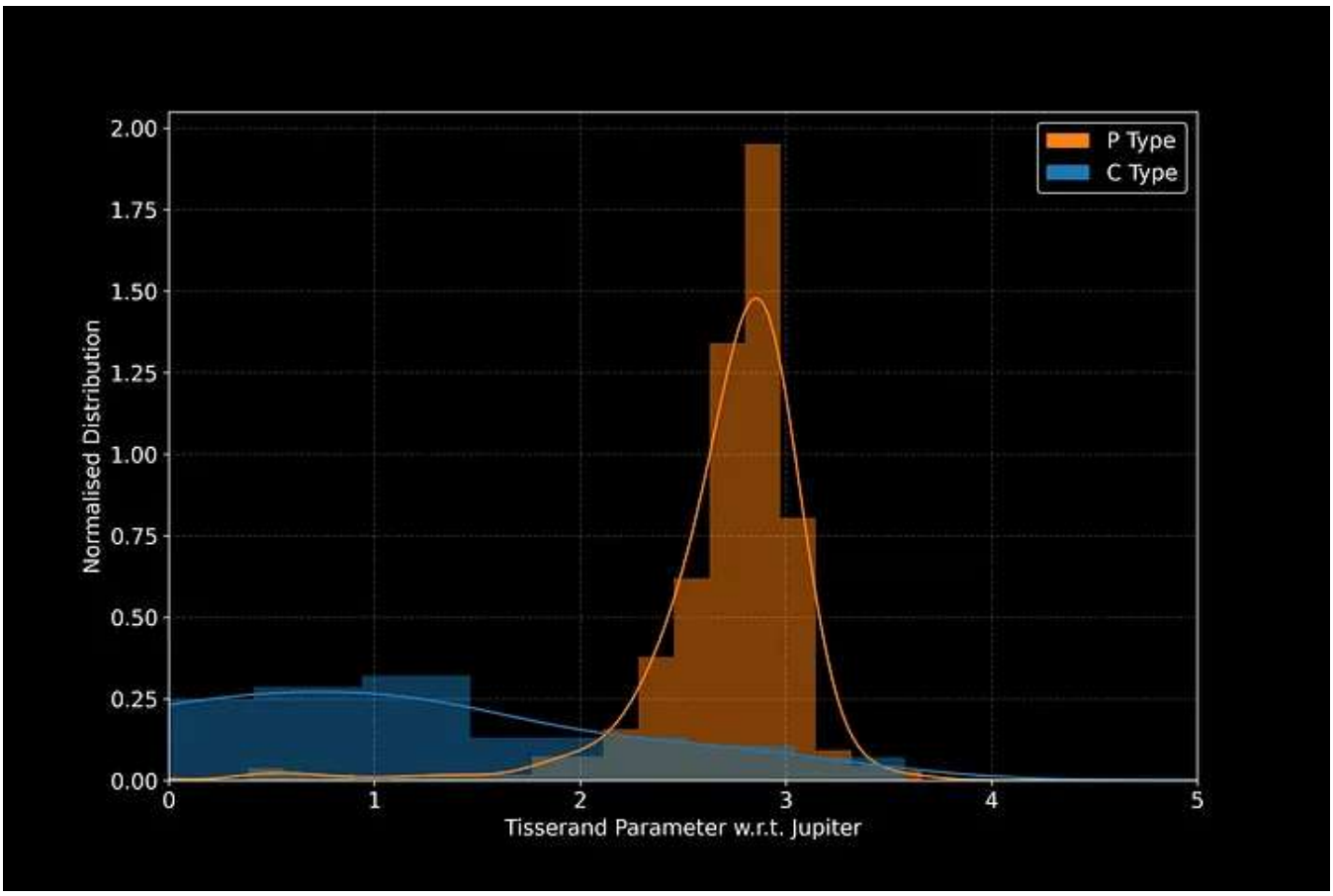
Part 7/9

In a final step, we can visualise the Tisserand parameter results with a KDE. We can use the code snippets from last time to create a Tisserand parameter distribution …

Part 8/9

… as well as the plot itself. This is the advantage of proper coding and fulfilling PEP8 standards and commenting: One can use code fragments and functions for future purposes without *re-inventing the wheel*. Plotting functions can also be easily put in a generic way as a common function.

The figure below shows the resulting distribution. The normalised distribution is plotted vs. the Tisserand parameter between 0 and 5. The blue and orange histogram and KDE show the data of the C and P type comets, respectively. You can see that the P type distribution is quite narrow and concentrated between a value of 2 and 3, as the printed statistics indicated. So, P types are more likely a JFC and are more dynamically linked to Jupiter as C type comets. *What does this imply?* Well, Jupiter "captures" long period comets and alters their orbital elements in a way that they become part of the inner Solar System. However, the aphelion of these comets corresponds approximately with the semi-major axis of Jupiter. After a few hundred years an additional close encounter with the gas giant alters the orbit again.



Histogram and KDE distribution of the Tisserand parameter of P and C type comets. Credit: T. Albin

Conclusion & Outlook

A theory from 1896 helped us to reveal the dynamical nature of P type comets. This theory is build purely on orbit dynamics, with certain assumptions and without any

data at that time to proof it! Having persistent theories and functions appear to be "boring" if you compare it with the fast living machine learning and data science world. But deriving a persistent theory that lasts for decades is an intellectual achievement that allows us to understand certain space scientific concepts on a very deep level.

The Tisserand parameter is a multi-dimensional equation. The next session will be a supplementary tutorial, where we will try to visualise this formula to get a better feeling of the equation. Then, we will determine bias effects in the comets' population and then … we will get familiar with a particular comet: 67P/Churyumov–Gerasimenko. 67P is well studied and we will create amazing 3D animations to understand the shape and origin of comets.

Thomas

References

[1] Murray, C., & Dermott, S. (2000). *Solar System Dynamics*. Cambridge: Cambridge University Press. doi:10.1017/CBO9781139174817

[2] Tisserand, F.-F. (1896). *Traité de Mécanique Céleste IV*. Gauthier-Villars, Paris

[3] Completing the Inventory of the Solar System, Astronomical Society of the Pacific Conference Proceedings, volume 107, T.W. Rettig and J.M. Hahn, Eds., pp. 173–191.

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